

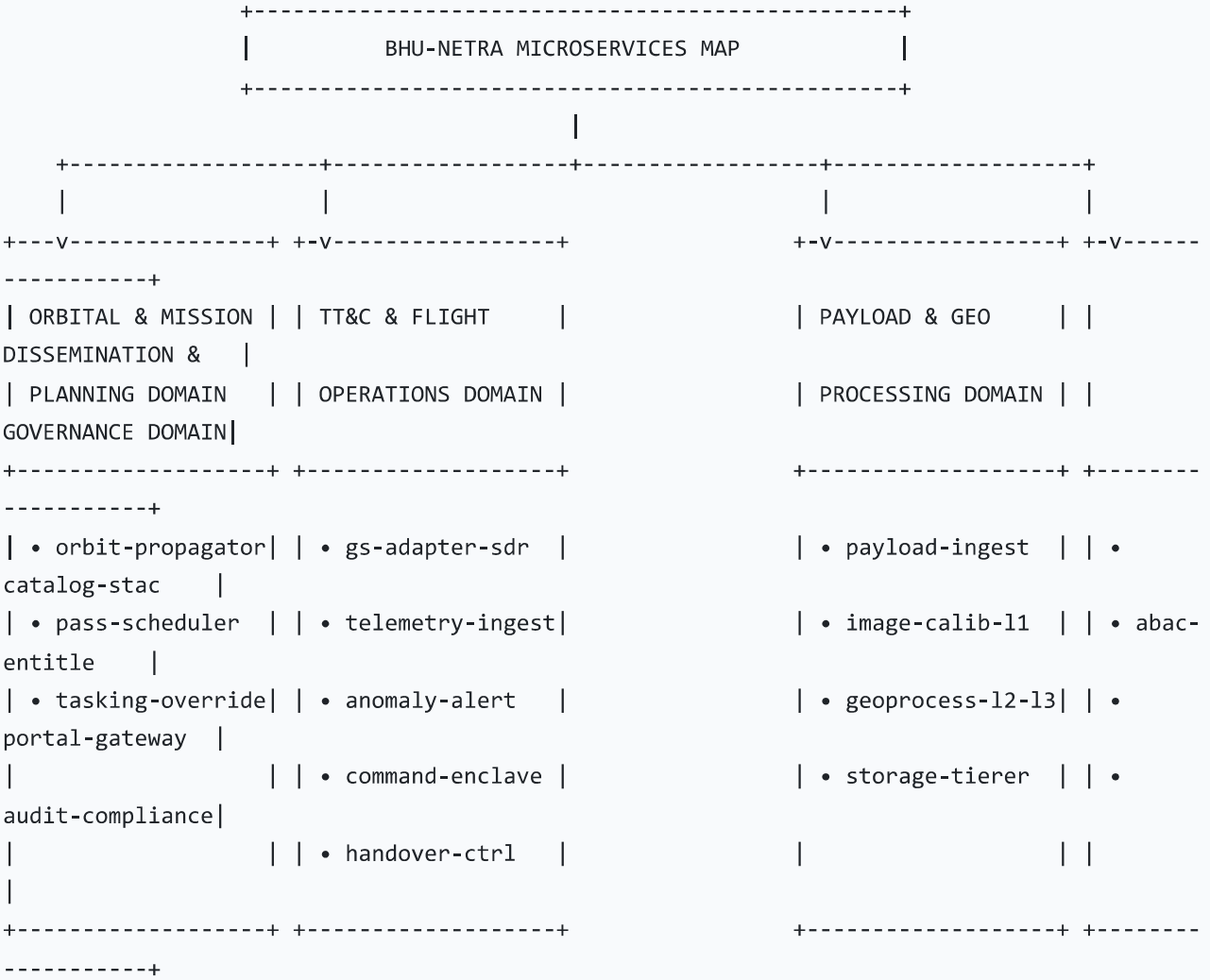
BHU-NETRA: Satellite Ground Segment & Mission Control Platform

Document 02: Microservices Architecture & Service Catalog

1. Architectural Overview & Domain Decomposition

BHU-NETRA is structured into **four core domain bounded contexts**:

1. **Orbital & Mission Planning Domain (OMPD)**: Pass scheduling, orbit propagation, emergency tasking.
2. **TT&C & Flight Operations Domain (TFOD)**: Baseband ingest, real-time telemetry, command enclave, GS handover.
3. **Payload & Geo-Processing Domain (PGPD)**: High-throughput ingestion, radiometric/geometric calibration, mosaic engine.
4. **Dissemination & Governance Domain (DGPD)**: STAC catalog, ABAC policy engine, public/government web portal.



2. Microservice Detailed Specifications

2.1 Orbital & Mission Planning Domain

Service 1: orbit-propagator-service

- **Primary Responsibility:** Solves satellite orbital mechanics (SGP4/SDP4 propagation), calculates TLE updates, predicts ground station visibility passes, elevation/azimuth acquisition loss times (AOS/LOS), and Doppler shifts.
- **Tech Stack:** Python / Java (Orekit library), C++ core for high-speed calculation.
- **Protocols:** gRPC for internal real-time calculations; REST for batch queries.
- **Database:** PostgreSQL (Satellite ephemeris, TLE records).

Service 2: `pass-scheduler-service`

- **Primary Responsibility:** Constraint-satisfaction engine (OR-Tools / OptaPlanner) that generates multi-satellite, multi-ground-station operational schedules considering battery levels, memory buffers, and sensor availability.
- **Protocols:** REST API, Kafka event emitter (`pass.scheduled`).
- **Database:** PostgreSQL (Pass schedules, conflict logs).

Service 3: `tasking-override-service`

- **Primary Responsibility:** Manages priority request queue (e.g. NDRF disaster response). Automatically recalculates schedules and revokes lower-priority imaging commands.
- **Protocols:** REST / gRPC, Kafka (`tasking.emergency_preempt`).

2.2 TT&C & Flight Operations Domain

Service 4: `gs-adapter-sdr-service`

- **Primary Responsibility:** Abstraction layer between physical Ground Station hardware and core software. Connects to legacy Baseband processors via VITA 49 / TCP streams and modern Software Defined Radios (GNU Radio / USRP X410).
- **Protocols:** VITA 49 (VITA Radio Transport over UDP), TCP sockets, C++ native bindings.
- **Database:** Redis (Live baseband link status, AGC levels, carrier lock state).

Service 5: `telemetry-ingestion-pipeline`

- **Primary Responsibility:** De-multiplexes CCSDS CADU/VCFrames, decodes Reed-Solomon / LDPC error correction, extracts telemetry parameters (Voltage, Thermal, AOCS), and calibrates raw counts to engineering units.
- **Protocols:** UDP stream, Apache Kafka ingestion (`telemetry.raw_frames` , `telemetry.calibrated`).
- **Database:** TimescaleDB (Time-series telemetry hypertable).

Service 6: `anomaly-alerting-service`

- **Primary Responsibility:** Real-time evaluation of telemetry parameter thresholds and ML anomaly models (Isolation Forest / LSTM autoencoders). Triggers instant mission alarms.
- **Protocols:** Kafka consumer, WebSocket stream to Mission Control Consoles, Webhooks (PagerDuty/SMS).
- **Database:** Redis (Active alarms cache), PostgreSQL (Historical anomaly events).

Service 7: `command-authorization-enclave`

- **Primary Responsibility:** Air-gapped, multi-sign-off state machine. Manages the 3-stage sign-off (Operator -> FDO -> Mission Director), integrates with FIPS 140-3 HSM for digital signatures, and generates encrypted CCSDS Telecommand frames.
- **Protocols:** gRPC (TLS 1.3 with mTLS & HSM PKCS#11 driver).
- **Database:** PostgreSQL (Immutable command authorization logs), Hardware Security Module (Private keys).

Service 8: `gs-handover-controller`

- **Primary Responsibility:** Coordinates real-time mid-pass handover between ground stations. Executes RF frequency lock transfer, telemetry stream sync, and command link migration.
- **Protocols:** gRPC, Kafka (`handover.initiated` , `handover.completed`).
- **Database:** Redis (Live handover state machine).

2.3 Payload & Geo-Processing Domain

Service 9: `payload-downlink-ingester`

- **Primary Responsibility:** Parallel high-speed ingestion of X-Band and Ka-Band raw sensor payloads (up to 10 Gbps per pass). Performs CRC verification and writes to Hot NVMe arrays.
- **Protocols:** VITA 49 / High-speed TCP socket, gRPC.
- **Database:** MinIO / S3 Object Storage (Raw Level-0 `.dat` files).

Service 10: `image-calibration-l1-service`

- **Primary Responsibility:** Performs dark current subtraction, flat-field correction, gain adjustment, and detector radiometric calibration.
- **Protocols:** Distributed worker queue (Ray Cluster / Celery / Airflow).

Service 11: `geoprocess-l2-l3-service`

- **Primary Responsibility:** Executes geometric orthorectification, DEM alignment (using Cartosat/SRTM DEMs), cloud masking, pan-sharpening, and COG (Cloud-Optimized GeoTIFF) generation.
- **Protocols:** GDAL / Orfeo Toolbox / PDAL bindings over Python/C++.
- **Database:** MinIO / S3 (Level-2 and Level-3 GeoTIFF products).

Service 12: `storage-tierer-service`

- **Primary Responsibility:** Automated lifecycle management of payload artifacts across Hot NVMe (0-7 days), Warm SAN (8-90 days), and Cold LTO Tape / Deep Archive (>90 days).

- **Protocols:** REST API, Cron schedule.

2.4 Dissemination & Governance Domain

Service 13: catalog-stac-service

- **Primary Responsibility:** Implements OGC SpatioTemporal Asset Catalog (STAC) API 1.0.0. Enables spatio-temporal spatial querying (BBOX, Point, Polygon, Date range, Sensor, Cloud cover %).
- **Protocols:** REST API (OpenAPI 3.0), GeoJSON.
- **Database:** PostgreSQL + PostGIS (Spatial indexing on geometry column).

Service 14: abac-entitlement-service

- **Primary Responsibility:** Attribute-Based Access Control (ABAC) engine. Evaluates user clearance (Gov Ministry vs Commercial Subscriber), spatial boundaries, classification policy, and spatial resolution limits before serving products.
- **Protocols:** gRPC / REST (Keycloak OAuth2 / OIDC JWT validation).
- **Database:** PostgreSQL (User policies, subscription contracts).

Service 15: portal-gateway-service

- **Primary Responsibility:** API Gateway providing public and G2G web portal backends. Integrates GeoServer for WMS/WMTS map tile rendering and image downloads.
- **Protocols:** REST, OGC WMS/WMTS/WFS.

Service 16: audit-compliance-logger

- **Primary Responsibility:** Centralized, tamper-evident audit ledger capturing every telecommand approval, ground station handover, telemetry anomaly, and imagery download.
- **Protocols:** Kafka event consumer, immutable append-only ledger.
- **Database:** PostgreSQL append-only tables / Elasticsearch.

3. Inter-Service Communication Matrix

Source Service	Target Service	Interaction Pattern	Protocol / Payload	Purpose
pass-scheduler-service	gs-adapter-sdr-service	Asynchronous Event	Kafka (pass.scheduled)	Upload tracking schedule to GS antenna controller
gs-adapter-sdr-service	telemetry-ingestion-pipeline	Stream	UDP / VITA 49 Stream	Ingest raw telemetry byte stream
telemetry-ingestion-pipeline	anomaly-alerting-service	Event Stream	Kafka (telemetry.calibrated)	Evaluate thresholds in real-time
command-authorization-enclave	gs-adapter-sdr-service	Synchronous mTLS	gRPC	Send cryptographically signed command frame for uplink
gs-handover-controller	gs-adapter-sdr-service	Synchronous gRPC	gRPC	Switch command carrier lock between GS nodes
payload-downlink-ingester	image-calibration-l1-service	Task Queue	Celery / Ray Queue	Trigger Level-1 calibration on completed pass
geoprocess-12-13-service	catalog-stac-service	REST / Event	Kafka (product.published)	Register COG metadata in PostGIS STAC Catalog
portal-gateway-service	abac-entitlement-service	Synchronous gRPC	gRPC	Verify subscriber license & classification before delivery